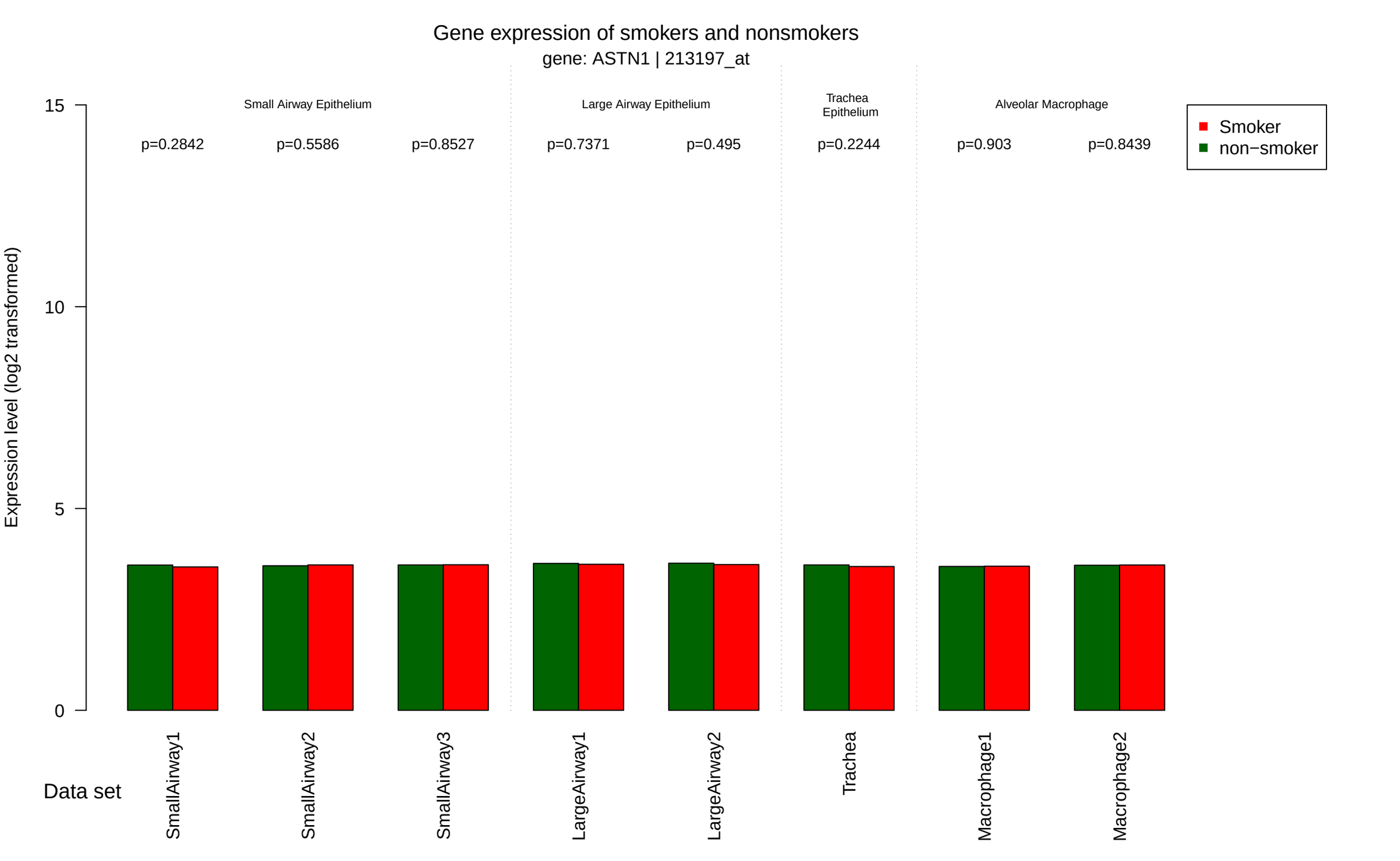
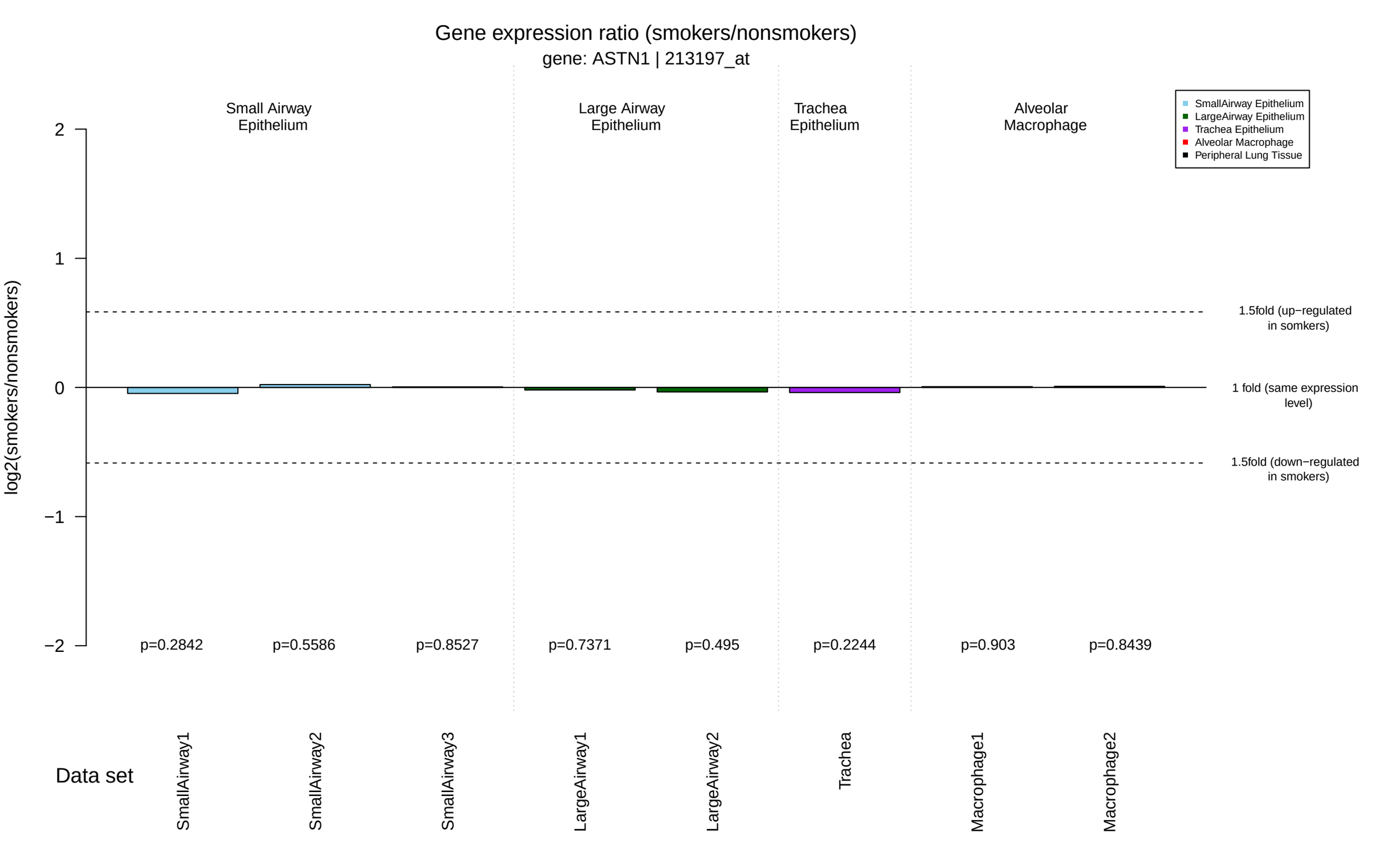
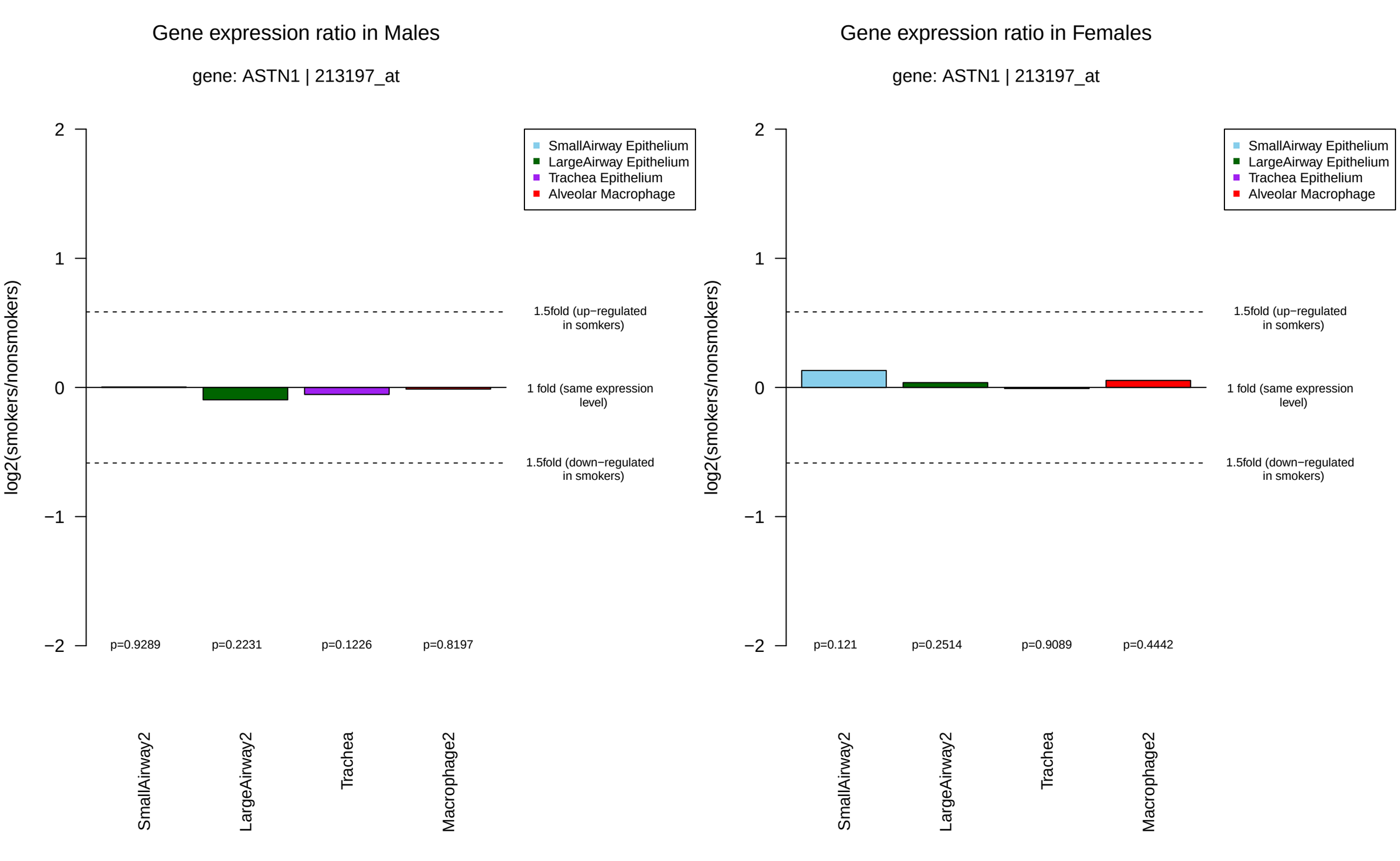
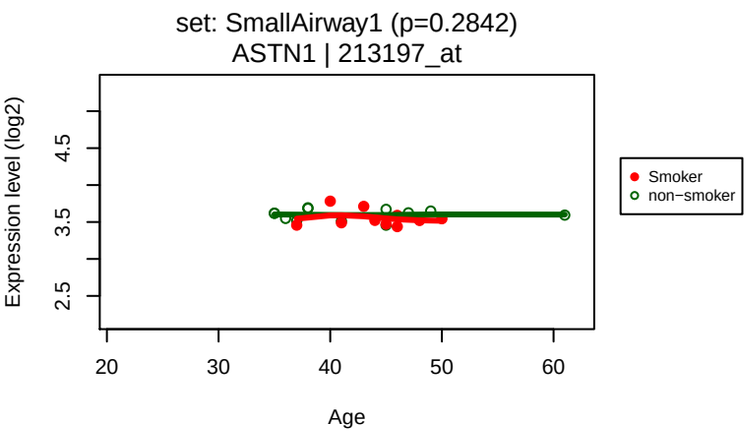
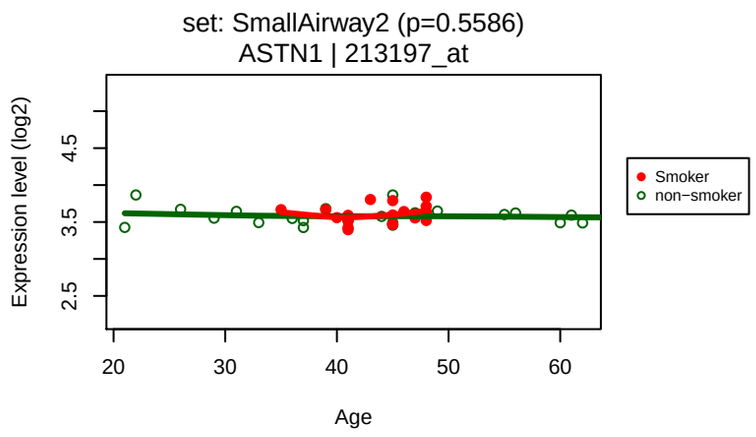
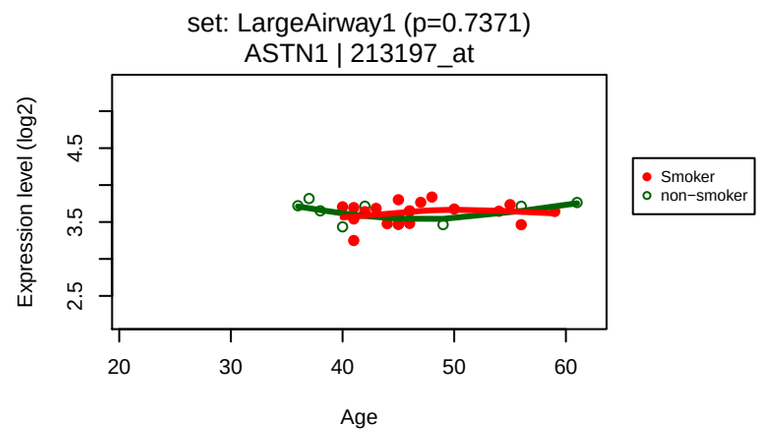

Gene expression of smokers and nonsmokers
gene: ASTN1 | 213197_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2842
p=0.5586
p=0.8527
p=0.7371
p=0.495
p=0.2244
p=0.903
p=0.8439
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

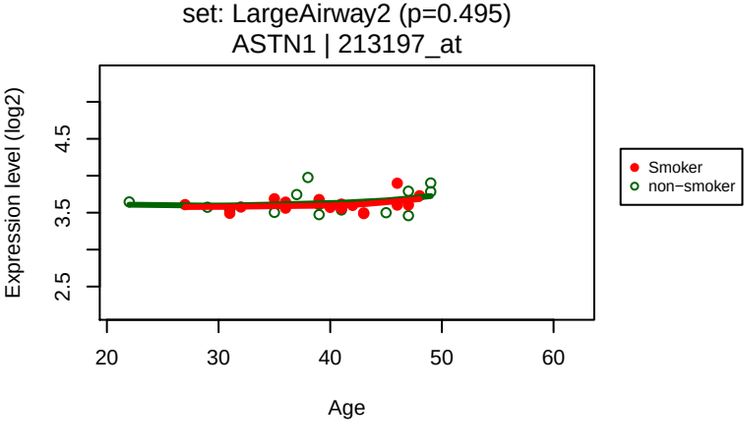

Gene expression ratio (smokers/nonsmokers)
gene: ASTN1 | 213197_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2842
p=0.5586
p=0.8527
p=0.7371
p=0.495
p=0.2244
p=0.903
p=0.8439
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

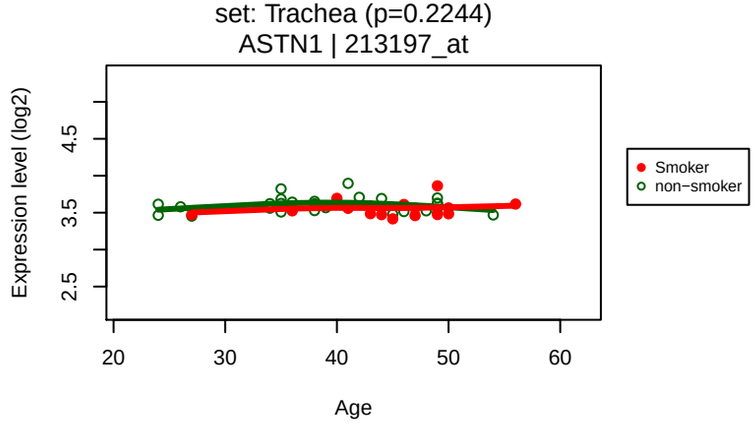

Gene expression ratio in Males
gene: ASTN1 | 213197_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9289
p=0.2231
p=0.1226
p=0.8197
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ASTN1 | 213197_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.121
p=0.2514
p=0.9089
p=0.4442
SmallAirway2
LargeAirway2
Trachea
Macrophage2

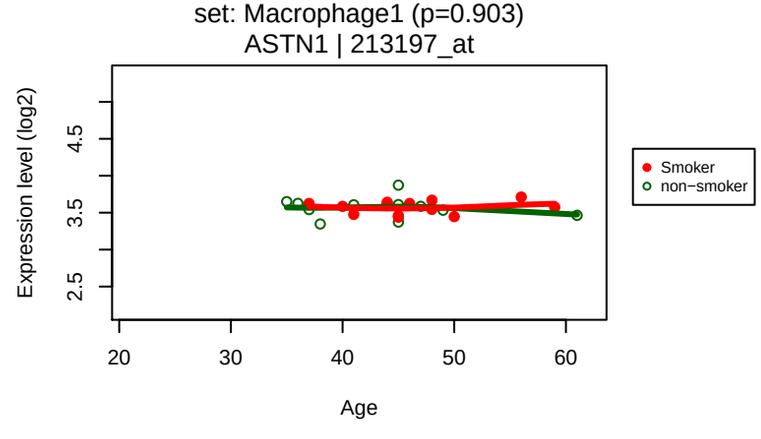

set: SmallAirway1 (p=0.2842)
ASTN1 | 213197_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker

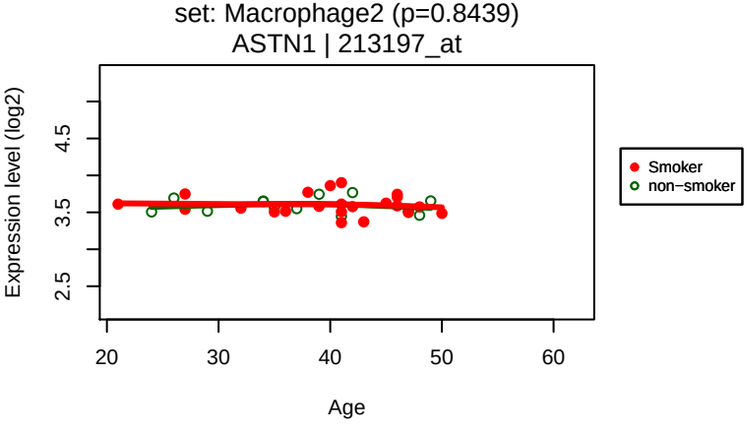

set: SmallAirway2 (p=0.5586)
ASTN1 | 213197_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker

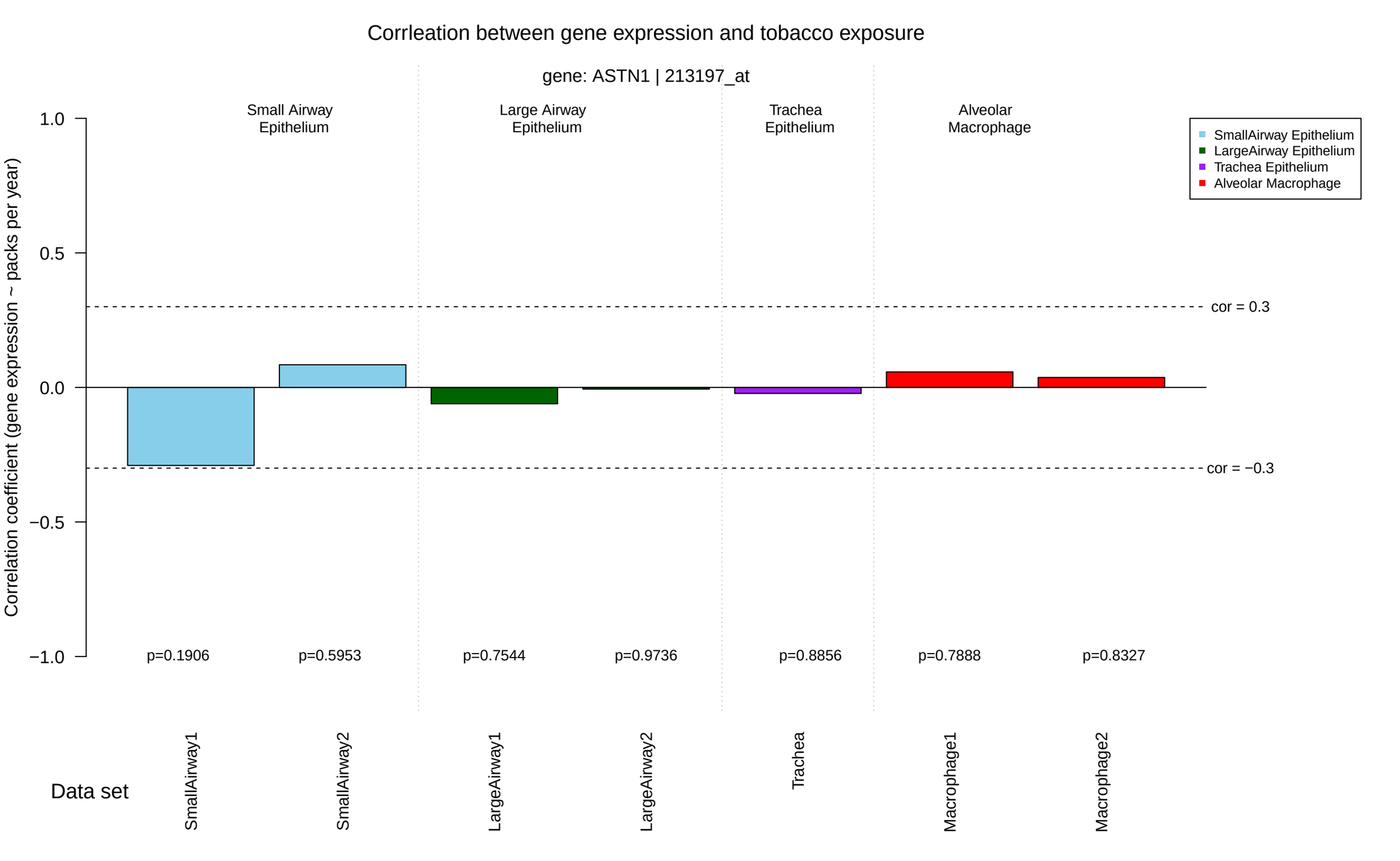

set: LargeAirway1 (p=0.7371)
ASTN1 | 213197_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.495)
ASTN1 | 213197_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.2244)
ASTN1 | 213197_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.903)
ASTN1 | 213197_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8439)
ASTN1 | 213197_at
Expression level (log2)
2.5
3.5
4.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ASTN1 | 213197_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1906
p=0.5953
p=0.7544
p=0.9736
p=0.8856
p=0.7888
p=0.8327
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2